

Trainings ▾

Leak Test

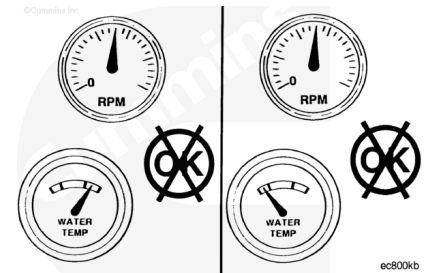
Some Signature™, ISX, and QSX engines use two engine coolant thermostats.

The engine thermostat(s) and thermostat seal(s) **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.

Use the following procedure to inspect the thermostat seal.

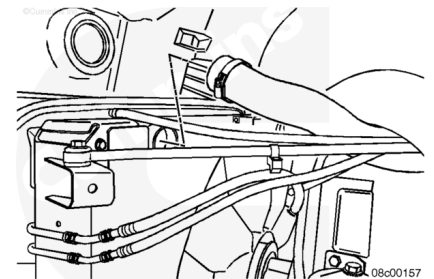
Refer to Procedure 008-016 in Section 8.

(/qs3/pubsys2/xml/en/procedures/10/10-008-016.html)



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the coolant to a level below the vent line in the thermostat housing.

Disconnect and plug the vent line hose. Refer to Procedure 008-017 in Section 8

(/qs3/pubsys2/xml/en/procedures/10/10-008-017.html).

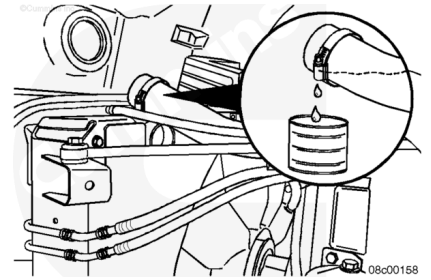
Remove the radiator hose from the radiator top tank.

The radiator hose **must** be at the point of overflowing before starting the engine. Small amounts of leakage will **not** be detected if the hose is **not** pre-filled.

Fill the radiator hose with coolant until coolant is just about ready to flow out of the hose.

Place a dry container under the end of the hose.

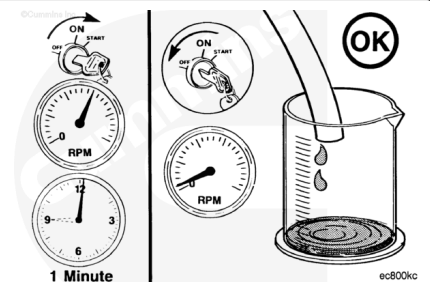
Note : If the original equipment manufacturer (OEM)-installed radiator hose is **not** long enough to safely perform this test, install a hose of the same size that is long enough to reach a remote, dry container used to collect coolant.



Operate the engine at rated rpm for 3 minutes.

Shut the engine OFF and measure the amount of coolant collected in the container.

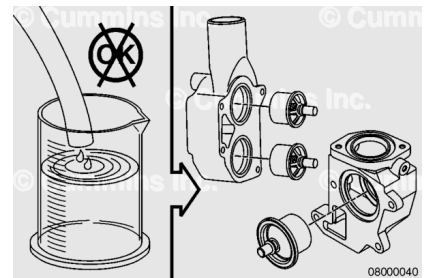
The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].



If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat(s) or thermostat seal(s) is leaking.

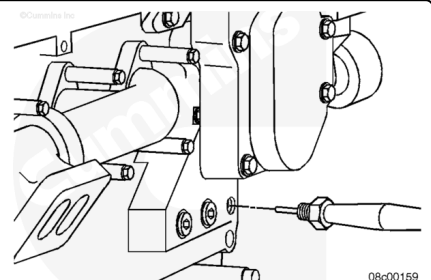
Remove the thermostat(s) and test operation as described in the Out-of-Chassis Opening Temperature Test in this procedure.

Refer to Procedure 008-016 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-016.html>) for thermostat seal inspection.



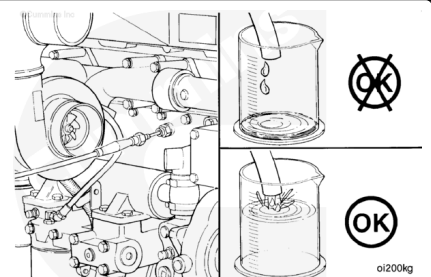
In-Chassis Opening Temperature Test

Remove the coolant temperature sensor and install a thermocouple or temperature gauge, which is known to be accurate, on the engine side of the thermostat housing.



Complete the following test in-chassis to test the thermostat(s) opening temperature.

Start the engine and monitor the water temperature and the container.



Thermostat Initial Opening Temperature		
celsius		fahrenheit
81	MIN	178
83	MAX	182

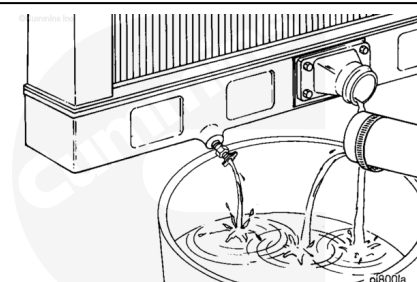
Shut the engine OFF when the coolant starts to flow.

If coolant does **not** start flowing into the container during the initial opening temperature range, the thermostat(s) **must** be replaced.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



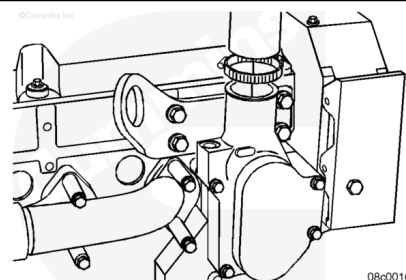
⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>) in Section 8.

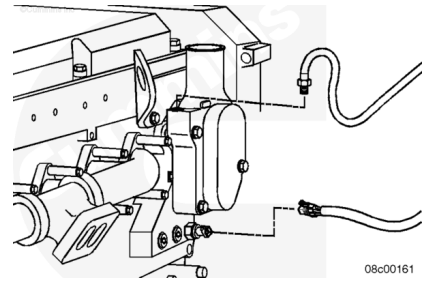
Remove

Remove the upper radiator hose from the thermostat housing cover.



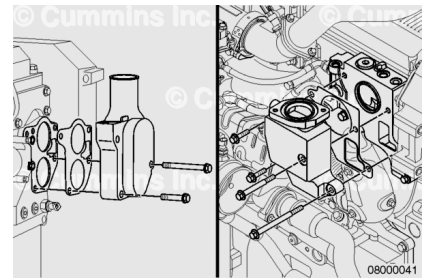
Disconnect the vent line.

Disconnect the coolant sensor connector.



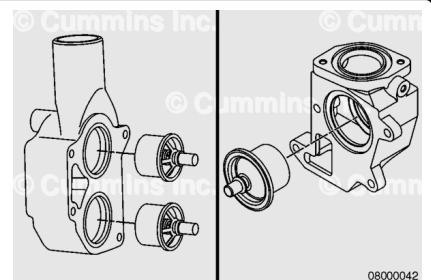
Remove the four capscrews that attach the thermostat housing cover to the thermostat housing.

Remove the thermostat housing cover and gasket from the thermostat housing. Discard the gasket.



Remove the thermostat and thermostat seal from the thermostat housing cover. Discard the thermostat seal. Refer to Procedure 008-016 in Section 8.

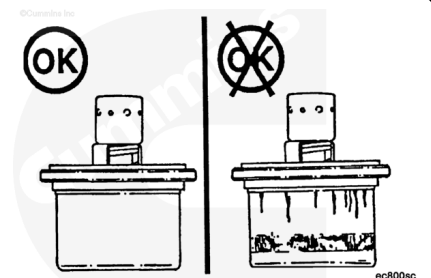
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Inspect for Reuse

Out-of-Chassis Opening Temperature Test

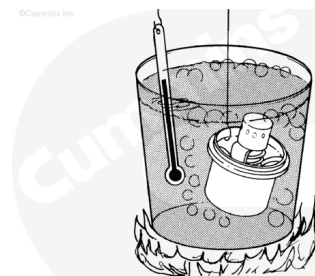
Inspect the thermostat for damage.



Suspend the thermostat and a 100°C [212°F] thermometer into a container of water. Do **not** allow the thermostat or the thermometer to touch the sides of the container.

Heat the water.

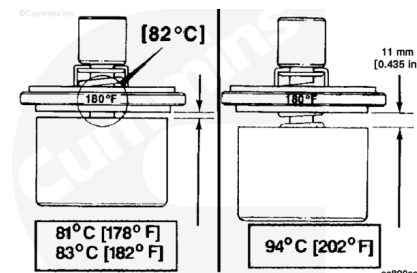
Record the temperatures at which the thermostat begins to open and when fully open.



The nominal operating temperature is stamped on the thermostat.

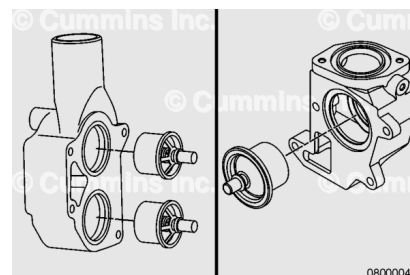
- The thermostat **must** begin to open within 1°C or 2°F of nominal temperature.
- The thermostat **must** be fully open to at least 11 mm [0.435 in] within 12°C or 22°F above nominal temperature.

Replace the thermostat if it does **not** operate as described.



Install

Install the thermostat(s) into the thermostat housing cover by pushing on the outer rims.



Install the thermostat housing cover, gasket, and four capscrews on the cylinder head.

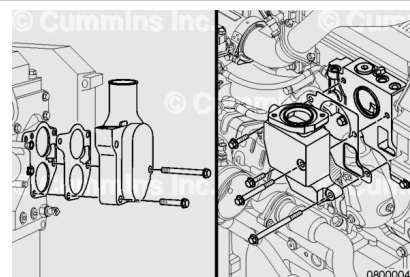
Tighten the capscrews to the values shown.

Torque Value:

Short Capscrews 34 n•m [25 ft-lb]

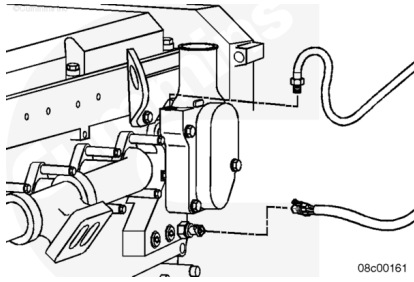
Torque Value:

Long Capscrews 56 n•m [41 ft-lb]



Connect the vent line. See equipment manufacturer service information.

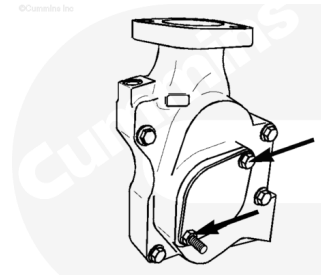
Connect the coolant sensor connector.



Industrial engines use a thermostat housing cover with either a cover plate or a water connection.

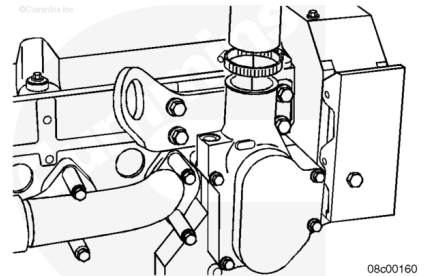
If removed, tighten the capscrews to the following specification.

Torque Value: 28 n•m [20 ft-lb]



Install the upper radiator hose. Tighten the hose clamp.

Torque Value: 5 n•m [40 in-lb]



Finishing Steps

- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Operate the engine until the coolant temperature reaches 70°C [160°F] and check for leaks.

